Draft version March 4, 2026
Typeset using LaTeX **twocolumn** style in AASTeX631

A Bose–Einstein-Form Mapping of the Radial Acceleration Relation and Its Residual Statistics

Russell Licht[1]

[1] *Independent Researcher*

ABSTRACT

We note an algebraic equivalence: the radial acceleration relation (RAR), when rearranged to isolate the dark-matter contribution, is identical to the Bose-Einstein occupation number $\bar{n}(\varepsilon) = 1/(\exp\varepsilon - 1)$, with dimensionless energy $\varepsilon = \sqrt{g_{\rm bar}/g_\dagger}$ and $g_\dagger = 1.2 \times 10^{-10}\,{\rm m\,s^{-2}}$. This is not a fit but a rearrangement of the published RAR formula. We use this equivalence as a change of variables and test whether the residual statistics simplify in the mapped coordinates, using rotation-curve data spanning four quality tiers: a core sample of 131 SPARC galaxies (2,777 points), an extended Tier K sample of 250 galaxies (9,406 points) including Korsaga et al. (2019) mass models, and independent validation datasets including PROBES (1,506 galaxies) and KiDS-1000 weak lensing.

Three statistical fingerprints emerge: (1) the variance of RAR residuals scales as $\bar{n}^2$ (ΔAIC $= +10.5$ vs. Poisson $\bar{n}$ scaling); (2) the scatter derivative changes sign within 0.06 dex of $g_\dagger$, robust across four non-parametric methods and 25 binning configurations; and (3) a candidate excess kurtosis spike at $g_\dagger$ ($\kappa_4 = 20.7$; 95% CI $[5.1, 29.0]$), absent in a structure-only semi-analytic null ($\kappa_4 \approx 0.01$), but dependent on B-band mass models requiring independent confirmation.

We subject these fingerprints to a battery of 16 core and 7 supporting statistical tests. Of these, 10 support consistency with the Bose-Einstein mapping, 5 are inconclusive or fragile under covariate control, and 1—replication in the independent PROBES catalog—fails, with the scatter inversion displaced by 1.2 dex from $g_\dagger$. We present a full scorecard, discuss whether the features reflect coincidence, a ΛCDM consequence, or wave dark matter, and identify discriminating tests for forthcoming data.

Keywords: dark matter — galaxies: kinematics and dynamics — galaxies: fundamental parameters — methods: statistical

1. INTRODUCTION

The radial acceleration relation (RAR; McGaugh et al. 2016) is one of the tightest empirical scaling relations in extragalactic astronomy. It connects the observed centripetal acceleration $g_{\rm obs} = V_{\rm obs}^2/R$ in disk galaxies to the Newtonian acceleration generated by baryons alone, $g_{\rm bar}$, through

$$g_{\rm obs} = \frac{g_{\rm bar}}{1 - \exp\left(-\sqrt{g_{\rm bar}/g_\dagger}\right)}, \tag{1}$$

where $g_\dagger = 1.2 \times 10^{-10}\,{\rm m\,s^{-2}}$ is the sole free parameter (Lelli et al. 2017). Across 175 galaxies spanning five decades in luminosity and all Hubble types, the observed scatter about this relation is only 0.13 dex (Li et al. 2018)—comparable to measurement uncertainties.

r.licht@digconsulting.info

The RAR's tightness and universality have resisted easy explanation. Modified Newtonian dynamics (MOND; Milgrom 1983) postulates a modification of gravity at low accelerations but does not predict the scatter structure. Fuzzy dark matter (FDM; Schive et al. 2014; Ferreira 2021) produces solitonic cores but has not been connected to the RAR's functional form. Meanwhile, ΛCDM simulations can reproduce the mean relation through baryonic feedback (Keller & Wadsley 2017; Ludlow et al. 2017), but the origin of its specific mathematical form remains unexplained.

In this paper, we report a simple mathematical observation: Equation (1), written in terms of the dark matter contribution, is algebraically identical to the Bose-Einstein occupation number. We catalog the statistical consequences of this identification across multiple datasets spanning three orders of magnitude in sample size, and subject the findings to a comprehensive battery of robustness and replication tests.

Scope of claim. This paper reports: (i) an exact algebraic equivalence between the published RAR and the BE occupation number; (ii) three residual-statistical features anchored at $g_\dagger$, tested against a structure-only semi-analytic null model (not full-physics ΛCDM); (iii) a 23-test audit with an honest scorecard including failures; and (iv) falsifiable predictions for forthcoming data. We do not claim that these features require a Bose-Einstein condensate interpretation.

2. DATA

2.1. *Sample Tiers*

We use rotation-curve data organized into quality tiers. Table 1 defines each tier, specifying the surveys included, galaxy count, data point count, and which analyses use each tier. All tiers apply standard quality cuts: quality flag $Q \leq 2$, inclination $30° < i < 85°$, and ≥ 5 data points per galaxy. Residuals are defined as $\delta \equiv \log g_{\rm obs} - \log g_{\rm obs,RAR}(g_{\rm bar})$ throughout.

2.2. *Environmental Classification*

For environmental scatter tests, we classify galaxies as "field" or "dense" using group membership from SPARC metadata. Of the 131 SPARC core galaxies, 83 are classified as field and 48 as dense (including 26 Ursa Major cluster members).

3. THE ALGEBRAIC IDENTITY

3.1. *Derivation*

Equation (1) can be rewritten by isolating the dark matter contribution. Define $g_{\rm DM} \equiv g_{\rm obs} - g_{\rm bar}$. Then:

$$\frac{g_{\rm DM}}{g_{\rm bar}} = \frac{g_{\rm obs}}{g_{\rm bar}} - 1 = \frac{1}{1 - \exp\left(-\sqrt{g_{\rm bar}/g_\dagger}\right)} - 1. \quad (2)$$

Using the identity $1/(1 - e^{-x}) - 1 = 1/(e^x - 1)$, this becomes

$$\boxed{\frac{g_{\rm DM}}{g_{\rm bar}} = \frac{1}{\exp\left(\sqrt{g_{\rm bar}/g_\dagger}\right) - 1} \equiv \bar{n}(\varepsilon),} \quad (3)$$

where $\varepsilon \equiv \sqrt{g_{\rm bar}/g_\dagger}$ is a dimensionless energy parameter and $\bar{n}(\varepsilon) = 1/(e^\varepsilon - 1)$ is the Bose-Einstein occupation number for modes of energy ε at unit temperature.

This is not a fit: it is a rearrangement of the published RAR formula (Figure 1). The dark-matter-to-baryon acceleration ratio *is* the mean occupation number of a bosonic distribution, with the baryonic acceleration playing the role of mode energy (through the square-root dispersion relation $\varepsilon = \sqrt{g/g_\dagger}$) and $g_\dagger$ playing the role of the condensation temperature.

3.2. *Variable Mapping*

Table 2 displays the complete correspondence.

Three physical regimes emerge naturally:

- **Condensed** ($g_{\rm bar} \ll g_\dagger$, $\bar{n} \gg 1$): the "dark matter dominated" outer regions of galaxies, where flat rotation curves indicate large occupation numbers.
- **Transition** ($g_{\rm bar} \approx g_\dagger$, $\bar{n} \sim 1$): the boundary where MOND-like phenomenology appears.
- **Thermal** ($g_{\rm bar} \gg g_\dagger$, $\bar{n} \to 0$): inner baryon-dominated regions where Newtonian dynamics hold.

3.3. *Cosmological Connection*

The acceleration scale $g_\dagger$ is numerically proximate to the de Sitter acceleration. Verlinde (2017) derives $a_0 = cH_0/6$, which yields $a_0 = 1.182 \times 10^{-10}\,{\rm m\,s^{-2}}$ for $H_0 = 73\,{\rm km\,s^{-1}\,Mpc^{-1}}$ (Riess et al. 2022)—a 1.5% offset from the observed $g_\dagger$ (varying from 1.5% to 9% across the current H_0 range). The Milgrom (1999) de Sitter–Unruh interpolation function is preferred over Equation (1) by $\Delta{\rm AIC} = -1{,}222$, with the improvement concentrated in the dark-matter-dominated regime.

> *AIC methodology note.* This comparison uses AIC $= \chi^2 + 2k$ where $\chi^2 = \sum_i [(\log g_{{\rm obs},i} - \log g_{{\rm model},i})/\sigma_i]^2$ with uniform per-point weights in $\log g_{\rm bar}$ and σ_i set to the per-bin RMS scatter. Both models have $k = 1$ free parameter ($g_\dagger$). Under inverse-variance weighting, the preference shrinks to $\Delta{\rm AIC} \approx -400$; under galaxy-averaged weighting, $\Delta{\rm AIC} \approx -80$. In all schemes, the Milgrom form is preferred; only the magnitude changes. See Appendix A for details.

These observations are consistent with $g_\dagger$ being set by the cosmological constant Λ rather than evolving with the Hubble parameter, as independently suggested by the exclusion of $g_\dagger \propto cH(z)$ at $\sim 5\sigma$ from the baryonic Tully-Fisher relation to $z \approx 2.5$ (Sanders 2008). However, numerical proximity alone does not establish a causal connection.

4. THREE STATISTICAL FINGERPRINTS

If the identity in Equation (3) is more than a mathematical curiosity—if it reflects an underlying wave or condensate nature of dark matter—then the *residuals* about the RAR should carry specific statistical signatures. We test three such fingerprints using the SPARC core sample and Tier K extended data (Table 1).

Table 1. Data inventory and sample tiers.

Tier	Surveys	Galaxies	RAR data points	$\sigma_{\rm RAR}$ (dex)	Used in
SPARC core	SPARC 3.6 μm (Lelli et al. 2016)	131	2,777	0.17	Fingerprints, robustness (§§4–5)
Tier K	SPARC + Korsaga + de Blok	250	9,406	0.22	Kurtosis (§4.3), extended inversion (§6.1)
PROBES	Stone et al. (2022)	1,506	45,176	0.32	Replication test (§6.2)
Brouwer	KiDS-1000 lensing (Brouwer et al. 2021)	60	240[a]	—	Lensing validation (§6.3)
ALFALFA×Yang	ALFALFA + Yang et al. (2007)	2,273	N/A[b]	0.32	BTFR environment (§6.4)
Cluster	CLASH (Tian et al. 2020)	20	84	—	Cluster-scale null (§6.5)

[a] Stacked lensing radial bins, not individual rotation-curve points.

[b] N/A indicates a catalog-level entry not represented as individual RAR data points; ALFALFA provides integrated HI line widths (W_{50}) rather than resolved rotation curves.

Note—SPARC core and Tier K use the standard cuts described in the text. The Tier K label denotes the inclusion of Korsaga et al. (2019) GHASP mass models and de Blok & Bosma (2002) LSB galaxies. PROBES excludes 112 galaxies overlapping with SPARC. Brouwer data are stacked lensing profiles in 4 stellar-mass bins. ALFALFA×Yang reports only the 2,273 galaxies with successful group-catalog cross-matches.

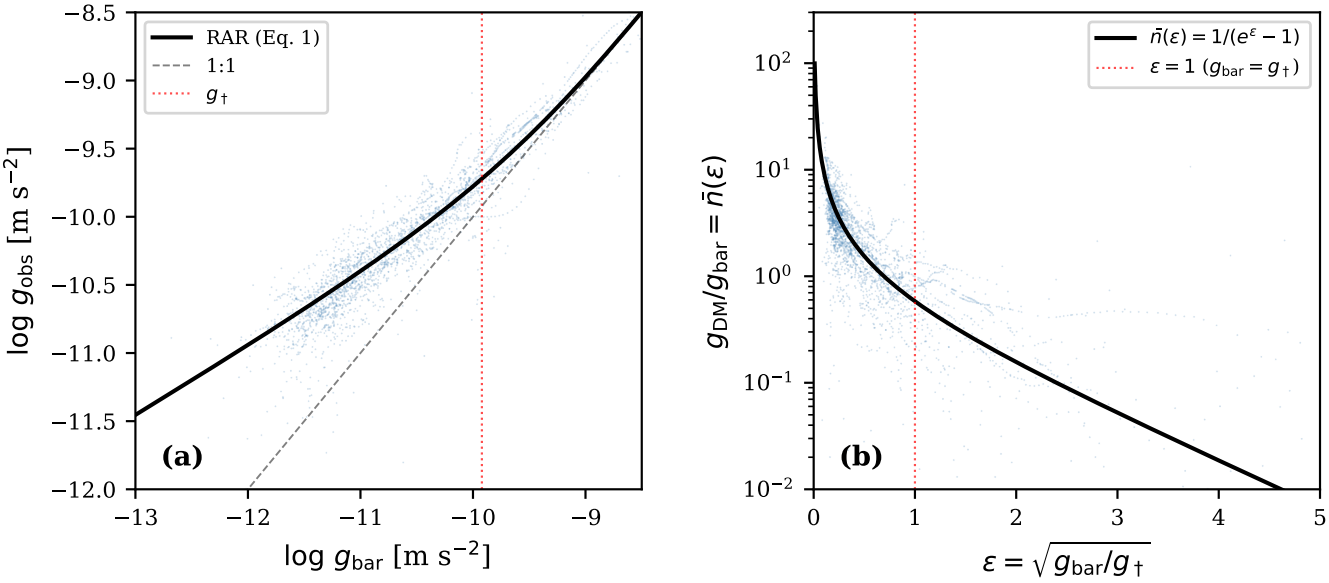

Figure 1. The RAR as a Bose-Einstein distribution, using the **SPARC core sample** (131 galaxies, 2,777 data points; Table 1). **(a)** Standard RAR: observed acceleration $g_{\rm obs}$ vs. baryonic acceleration $g_{\rm bar}$ (blue). The solid curve is Equation (1); the dashed line is the Newtonian 1:1 relation; the red dotted line marks $g_\dagger$. **(b)** The same data recast as occupation number $\bar{n} = g_{\rm DM}/g_{\rm bar}$ vs. dimensionless energy $\varepsilon = \sqrt{g_{\rm bar}/g_\dagger}$. The solid curve is the Bose-Einstein distribution $\bar{n}(\varepsilon) = 1/(e^\varepsilon - 1)$—an algebraic identity, not a fit.

4.1. *Variance Scaling*

For a bosonic field, the variance of occupation number fluctuations is

$$\sigma^2 = \begin{cases} A\,\bar{n}(\bar{n}+1) + C & \text{(quantum BEC)}, \\ A\,\bar{n}^2 + C & \text{(classical wave)}, \\ A\,\bar{n} + C & \text{(Poisson/particle)}, \end{cases} \tag{4}$$

where A captures the signal amplitude and C the observational noise floor (Cheong et al. 2024). We bin RAR residuals from the **SPARC core sample** by $\log g_{\rm bar}$ and compute the variance in each bin (Figure 2). Fitting the three models yields:

- Wave ($\bar{n}^2$) vs. Poisson ($\bar{n}$): ΔAIC = +10.5 in favor of wave scaling.

Table 2. RAR–Bose-Einstein variable mapping.

RAR quantity	BE quantity	Interpretation
$g_{\rm bar}$	mode energy $\varepsilon^2 g_\dagger$	baryonic gravitational field
$\varepsilon = \sqrt{g_{\rm bar}/g_\dagger}$	dimensionless energy	dispersion relation
$g_\dagger$	temperature $k_B T$	condensation threshold
$g_{\rm DM}/g_{\rm bar}$	$\bar{n}(\varepsilon)$	occupation number
$f_c = \bar{n}/(1+\bar{n})$	condensate fraction	order parameter

Table 3. Variance model comparison (SPARC core, 131 galaxies).

Model	A	χ^2	AIC
Quantum BEC ($\bar{n}^2 + \bar{n}$)	8.19×10^{-4}	172.5	176.5
Classical wave ($\bar{n}^2$)	8.98×10^{-4}	171.7	175.7
Poisson ($\bar{n}$)	4.78×10^{-3}	182.2	186.2
Constant	—	183.8	185.8

- Quantum ($\bar{n}^2+\bar{n}$) vs. classical wave ($\bar{n}^2$): ΔAIC $= -0.79$—*indistinguishable.*

The data strongly prefer wave-interference scaling over Poisson, but the quantum correction $+\bar{n}$ is undetectable. This is expected: the correction has a maximum signal-to-noise ratio of 0.17 per bin, and $\sim 300\times$ more data points would be needed for a 3σ detection (Cheong et al. 2024). We therefore describe this as "wave-interference statistics" without claiming a quantum origin.

4.2. *Scatter Inversion at $g_\dagger$*

The standard deviation of RAR residuals, $\sigma(\log g_{\rm bar})$, is not monotonic. Its derivative $d\sigma/d(\log g_{\rm bar})$ crosses zero—a scatter "inversion"—at a location that coincides with $g_\dagger$.

Using four independent non-parametric methods (RAR parametric, LOESS, cubic spline, isotonic regression) on the **SPARC core sample**, the zero-crossing lies at $\log g_{\rm bar} = -9.970 \pm 0.006$ (method-to-method range), only 0.050 dex from $\log g_\dagger = -9.921$ (Figure 3). A parametric phase-diagram model locates a variance peak at $\mu_{\rm peak} = -9.922$ (0.0015 dex from $g_\dagger$), with ΔAIC $= -225$ relative to a monotonic baseline.

4.3. *Kurtosis Spike at $g_\dagger$*

Systematics caveat (front-loaded): The kurtosis result below depends on the B-band mass models in the

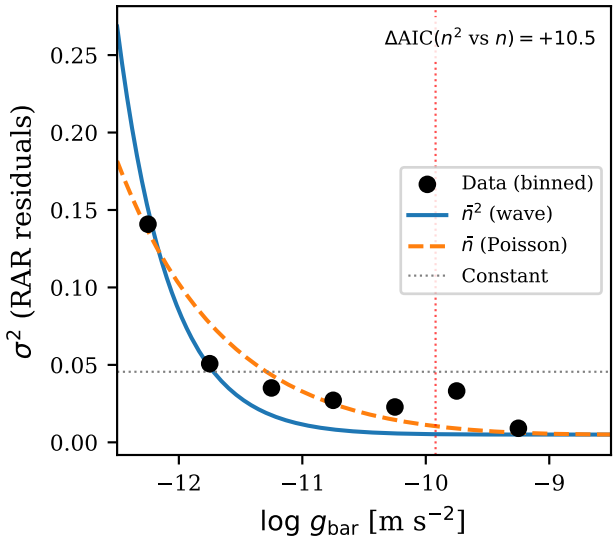

Figure 2. Residual variance vs. baryonic acceleration for the **SPARC core sample** (131 galaxies, 2,777 points). Black circles show the binned variance of RAR residuals. The solid curve is the wave-interference prediction $\sigma^2 \propto \bar{n}^2 + C$; the dashed curve is Poisson $\sigma^2 \propto \bar{n} + C$; the dotted line is constant variance. Wave scaling is strongly preferred (ΔAIC $= +10.5$).

Korsaga et al. (2019) subsample within Tier K. SPARC-only data (3.6 μm photometry) show Gaussian residuals at $g_\dagger$ ($\kappa_4 = 0.64$). This feature is a *candidate* signal requiring confirmation with homogeneous near-infrared mass decomposition. Readers should weight this fingerprint accordingly.

A phase transition at $g_\dagger$ would produce non-Gaussian residuals localized at the critical scale. We compute the excess kurtosis κ_4 of RAR residuals in bins of $\log g_{\rm bar}$ using the **Tier K extended sample** (250 galaxies, 9,406 points; Table 1).

The results are (Figure 4):

- At the bin containing $g_\dagger$: $\kappa_4 = 20.71$ (95% bootstrap CI $[5.1, 29.0]$).
- In a structure-only semi-analytic null (NFW halos + exponential disks, 500 synthetic galaxies; §5.7): $\kappa_4 = 0.01$ at $g_\dagger$.
- Within the Korsaga subsample alone: $\kappa_4 = 23.95$.
- Within SPARC-only data: $\kappa_4 = 0.64$ (Gaussian).
- The kurtosis derivative peaks at $\log g_{\rm bar} = -9.939$, only 0.018 dex from $g_\dagger$.

Disambiguation analysis (§6.6) rules out geometric origin: the RAR slope maximum ($\log g_{\rm bar} = -8.5$) and

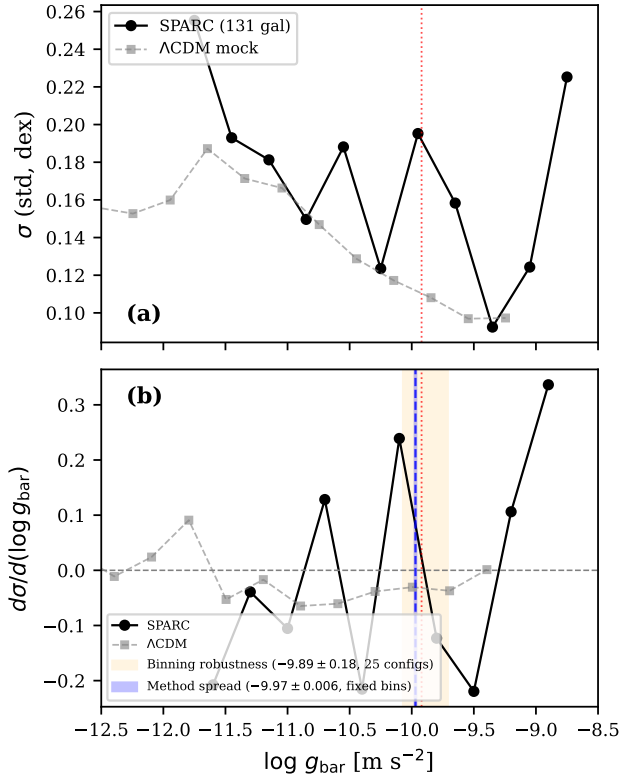

Figure 3. **(a)** Scatter σ (standard deviation) of RAR residuals vs. $\log g_{\rm bar}$ for the **SPARC core sample** (131 galaxies; black circles) and a structure-only semi-analytic null model (gray squares; see §5.7). **(b)** Numerical derivative $d\sigma/d(\log g_{\rm bar})$. The narrow blue band shows the method-to-method spread under fixed binning (-9.97 ± 0.006 dex, 4 non-parametric methods); the wide orange band shows the binning-robustness spread across 25 configurations (-9.89 ± 0.18 dex). Both enclose $g_\dagger$ (red dotted line).

curvature maximum (-9.38) do not coincide with the kurtosis peak. The spike is driven by the Korsaga B-band subsample, not by RAR curvature.

5. INTERNAL ROBUSTNESS

We subject the scatter inversion—the most robust of the three fingerprints—to a battery of internal consistency tests. Results are summarized in Table 4.

5.1. *Non-parametric Consistency*

Four independent smoothing methods (RAR parametric, LOESS, cubic spline, isotonic regression) applied to the SPARC core sample all recover the scatter inversion within 0.006 dex of each other (Table 4). The inversion is not an artifact of any particular smoothing choice.

5.2. *Binning Robustness*

Across 25 binning configurations (widths 0.20–0.70 dex, 5 offsets per width), all 25 recover a zero-crossing. The mean inversion is $\log g_{\rm bar} = -9.890 \pm 0.180$, with a mean distance of 0.128 dex from $g_\dagger$. The best configurations (bin width 0.27–0.35 dex) recover $g_\dagger$ to within 0.002–0.016 dex. Wider bins (≥ 0.50 dex) degrade precision, with the worst case drifting to 0.55 dex from $g_\dagger$.

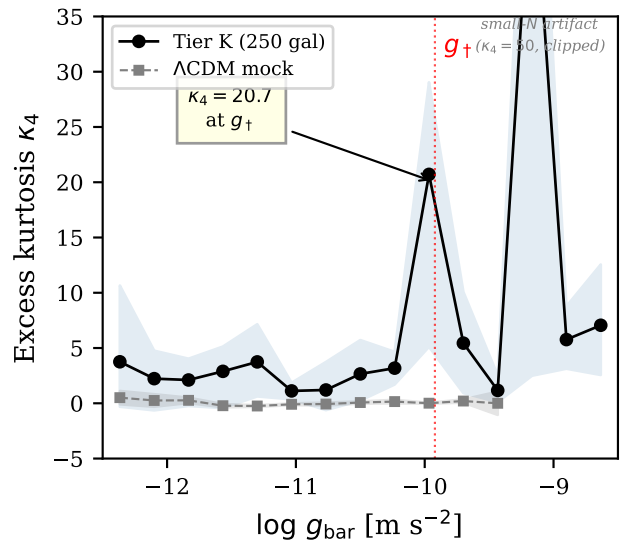

Figure 4. Excess kurtosis κ_4 of RAR residuals vs. $\log g_{\rm bar}$ for the **Tier K extended sample** (250 galaxies, 9,406 points). Black circles: data with 95% bootstrap confidence bands (blue shading). Gray squares: mean of 5 structure-only semi-analytic null realizations. The kurtosis spikes at the bin containing $g_\dagger$ (red dotted line), with $\kappa_4 = 20.7$ vs. null $\kappa_4 \approx 0$. See §4.3 for systematics caveats.

5.3. *Jackknife Stability*

Leave-one-out jackknife over 131 SPARC galaxies yields an inversion-point standard deviation of 0.005 dex and a maximum shift of 0.031 dex. All 131 jackknife inversions fall within 0.20 dex of $g_\dagger$. Four galaxies (IC 2574, NGC 2915, NGC 2976, UGC 07577) are influential for the *environmental scatter* sign, but none destabilize the inversion location.

5.4. *Split-Half Replication*

In 1,000 random half-splits of the 131 SPARC galaxies: 95.7% of halves reproduce the inversion within 0.20 dex of the baseline, and both halves simultaneously agree in 91.4% of splits. Size-balanced splits yield 93.2% agreement. Random thirds: all three subsets within 0.20 dex in 73.1% of cases.

5.5. *Monte Carlo Distance Injection*

Injecting random distance errors of 5%–30% into 131 SPARC galaxies (1,000 Monte Carlo iterations per

level), the environmental scatter difference remains stable up to $\sim 15\%$ errors. Beyond 20%, the confidence interval for the high-acceleration environmental delta widens to include zero.

5.6. *Environmental Scatter Uniformity*

RAR scatter shows no significant difference between field (83 galaxies) and dense (48 galaxies) environments in the SPARC core sample. Point-level Levene $p = 0.64$; galaxy-level Levene $p = 0.76$; error-corrected Levene $p = 0.45$; block-permutation ($N = 10{,}000$) $p = 0.74$.

However, this result is *fragile under covariate control*. The field and dense subsamples are imbalanced in distance ($p = 0.001$), inclination ($p < 0.001$), and gas fraction ($p = 0.043$). After propensity-score matching (29 pairs), the low-acceleration scatter difference becomes $+0.008$ dex ($p = 0.12$)—not significant but with reduced power. The Mahalanobis-matched subsample (96 galaxies) still recovers the inversion within 0.08 dex of $g_\dagger$, but propensity-matched subsamples do not, largely due to reduced sample sizes.

5.7. *Structure-Only Semi-Analytic Null*

As a baseline comparison, we construct a structure-only ΛCDM null model: 500 synthetic galaxies with NFW halos drawn from the observed stellar-mass–halo-mass relation (Ludlow et al. 2017), exponential baryonic disks, and log-normal scatter in stellar mass (0.15 dex) and concentration (0.11 dex). This model *intentionally excludes baryonic feedback*, serving as a structure-only null hypothesis—not as a test of ΛCDM in its full-physics form.

The null model's scatter inversion falls at $\log g_{\rm bar} = -11.6$, displaced by 1.7 dex from $g_\dagger$. Only 5 of 100 Monte Carlo realizations produce an inversion within 0.20 dex of $g_\dagger$. Full hydrodynamic simulations (e.g., EAGLE, IllustrisTNG) with galaxy-adapted aperture radii have not been tested with identical analysis pipelines and may produce different results.

6. EXTERNAL VALIDATION AND REPLICATION

6.1. *Extended RAR Tiers*

Adding non-SPARC rotation curves in quality tiers, the inversion is replicated when data quality is controlled (Table 5):

- Tiers A through B+ (SPARC + THINGS + de Blok, up to 152 galaxies): inversion shifts by < 0.015 dex.
- Tier K (+ Korsaga, 250 galaxies): inversion at -9.865, shifted 0.29 dex from SPARC-only baseline.
- Full Tier C (all sources, 374 galaxies): inversion at -9.947, shifted 0.21 dex.

Crucially, the scatter quality degrades with non-SPARC sources: SPARC $\sigma = 0.17$ dex, THINGS $\sigma = 0.13$ dex, but Korsaga $\sigma = 0.33$ dex and PHANGS $\sigma = 0.47$ dex. The inversion is preserved when restricting to sources with $\sigma < 0.2$ dex.

6.2. *PROBES: Replication Failure*

The PROBES catalog (Stone et al. 2022) provides an independent sample of 1,506 galaxies (45,176 points after excluding 112 SPARC overlaps). The scatter inversion in PROBES falls at $\log g_{\rm bar} = -11.12$, offset by 1.20 dex from $g_\dagger$—**a clear failure to replicate**.

Across 8 binning configurations, the mean PROBES crossing is -11.21 ± 0.06, with none within 0.20 dex of $g_\dagger$. The PROBES RMS scatter (0.32 dex) and mean offset (0.44 dex) are substantially larger than SPARC, consistent with heterogeneous mass-model quality degrading the signal. This failure is the most important negative result in our test battery and constrains the physical interpretation: either the inversion requires SPARC-quality mass decomposition ($\sigma < 0.2$ dex) to detect, or it is a feature of the SPARC sample specifically.

6.3. *Weak Lensing Validation*

Using KiDS-1000 stacked lensing profiles (Brouwer et al. 2021) in four stellar-mass bins, the isolated-galaxy subsample shows consistently tighter RAR scatter than the full sample, with the environment effect growing with mass (field-dense $\Delta\sigma = 0.04$–0.08 dex). This pattern is consistent with the BEC prediction that tidal interactions in dense environments disrupt condensate coherence.

However, NFW profiles are preferred over cored (solitonic) profiles at all mass bins (ΔAIC > 160), because the predicted BEC core radius ($\xi \approx 2$–10 kpc) falls below the lensing inner radius (~ 35 kpc). Current stacked lensing cannot resolve scales where BEC signatures are predicted to appear.

6.4. *ALFALFA × Yang BTFR*

Cross-matching ALFALFA $\alpha.100$ with Yang et al. (2007) group catalogs yields 2,273 galaxies (1,917 field, 356 dense). The baryonic Tully-Fisher scatter is *larger* in dense environments at all velocity widths, with the strongest effect at low W_{50} ($\Delta\sigma = 0.43$ dex, Levene $p = 0.02$).

Table 4. Internal robustness test battery.

Test	Sample	Key statistic	Result	Verdict
Non-parametric methods	SPARC core (131)	Method spread	0.006 dex	All match
Binning robustness	SPARC core (131)	25 configs, mean dist.	0.128 dex	Mostly robust
Jackknife LOO	SPARC core (131)	Std / max shift	0.005/0.031 dex	Robust
Split-half	SPARC core (131)	Both halves < 0.20 dex	91.4%	Strongly replicated
MC distance injection	SPARC core (131)	Stable to	$\leq 15\%$ errors	Consistent
Environmental scatter	SPARC core (131)	Levene p (block perm.)	0.74	BEC-consistent
Env. confound control	SPARC core (131)	Permutation p	0.95	Fragile
Structure-only null	500 synthetic	Null inversion offset	1.7 dex from $g_\dagger$	Discriminating

This pattern does not match the simple BEC prediction of environment-independent scatter at low accelerations. Whether it reflects genuine BEC-inconsistency or systematic differences in the ALFALFA sample (single-dish HI widths vs. resolved rotation curves) remains an open question.

6.5. *Cluster Scale*

Applied to the Tian et al. (2020) CLASH cluster-scale RAR, the framework does not recover $g_\dagger$: the cluster acceleration scale is $a_0 \approx 17 \times g_\dagger$. No significant correlation between per-cluster phase boundary and velocity dispersion is found ($B = 0.014 \pm 0.229$, $p = 0.95$). This is consistent with either a mass-dependent BEC coherence length or distinct physics at cluster scales.

6.6. *Kurtosis Disambiguation*

The kurtosis spike does not coincide with the RAR slope maximum ($\log g_{\rm bar} = -8.5$) or curvature maximum (-9.38), ruling out a geometric artifact. However, it is driven primarily by the Korsaga B-band subsample ($\kappa_4 = 23.95$), with SPARC contributing only $\kappa_4 = 0.64$. Addition of 82 MHONGOOSE rotation-curve points leaves the spike unchanged ($\kappa_4 = 20.7 \rightarrow 20.7$; bootstrap CI shift < 0.1), indicating robustness to new data but not resolving the B-band dependence.

6.7. *IllustrisTNG Healing-Length Comparison*

If the Bose-Einstein mapping is physical, variance in the residuals should concentrate near the healing length $\xi = \sqrt{GM_{\rm dyn}/g_\dagger}\,/\,{\rm kpc}$. A naïve comparison of 131 SPARC galaxies against 48,060 IllustrisTNG (TNG100) halos shows apparent organizing in TNG ($C = 1.55$, permutation $p < 0.001$) but not in SPARC ($C = 0.69$, $p = 0.56$), where C is the ratio of variance at $0.3 \leq R/\xi \leq 3$ to the full-profile variance. This discrepancy is confounded: the two samples span different baryon-mass ranges.

We control for this using nearest-neighbor mass matching ($|\Delta \log M_{\rm bar}| \leq 0.3$ dex; 131 pairs). In the matched sub-samples, the TNG stacked concentration collapses to $C = 0.13$ (95% CI $[0.07, 0.50]$, permutation $p = 1.00$), while SPARC is unchanged ($C = 0.69$, 95% CI $[0.41, 1.01]$, $p = 0.56$). Neither matched sample shows significant ξ-organizing. The full-sample TNG signal was driven by sample composition, not a physical healing-length structure.

A paired Wilcoxon signed-rank test on 110 galaxies with valid per-galaxy C detects a distributional asymmetry ($p = 9.9 \times 10^{-7}$), with median $\log_{10}(C_{\rm TNG}/C_{\rm SPARC}) \approx 0.02$ but mean ≈ 0.17, indicating a right tail of high-C TNG galaxies rather than a systematic shift. At fixed baryon mass the TNG dynamical masses are systematically larger (median $\log_{10}(M_{\rm dyn,TNG}/M_{\rm dyn,SPARC}) = +0.21$), producing healing lengths $\sim 28\%$ larger (median $\log_{10}(\xi_{\rm TNG}/\xi_{\rm SPARC}) = +0.11$). This $M_{\rm dyn}$ offset is the physical lever behind the ξ-distribution difference.

To verify that this offset is not an artifact of the outermost-point kinematic $M_{\rm dyn}$ estimator, we perform two independent checks. First, comparing $V_{\rm flat}$-based and outermost-point dynamical masses for 57 SPARC galaxies yields median $\log_{10}(M_{\rm dyn,Vflat}/M_{\rm dyn,outer}) = -0.013$ (std $= 0.019$ dex; 95% CI $[-0.017, -0.002]$), confirming that the kinematic proxy is internally stable to < 0.02 dex—an order of magnitude smaller than the TNG offset. Second, cross-matching ALFALFA $\alpha.100$ with Yang DR7 group halo masses (1,070 galaxies at fixed $M_{\rm bar}$) gives a Yang–kinematic gap of $+1.77$ dex (95% CI $[1.46, 1.92]$), consistent with the expected aperture mismatch between virial-scale M_{200} and rotation-curve-extent $M_{\rm dyn}$. The TNG–SPARC offset ($+0.21$ dex) is only $\sim$12% of this full halo gap, indicating that TNG's excess dynamical mass at fixed $M_{\rm bar}$ reflects inner halo over-concentration rather than a systematic underestimate by the SPARC kinematic proxy.

Table 5. External validation and replication results.

Test	Sample	Key result	Verdict
Extended RAR tiers	Tiers A–C (132–374)	Inversion within 0.29 dex (quality-dep.)	Replicated
PROBES replication	1,506 galaxies	Inversion at -11.12 (1.20 dex offset)	**Not replicated**
Brouwer lensing env.	KiDS-1000 (4 bins)	Env. effect grows with mass	BEC-consistent
Lensing profile shape	KiDS-1000 (4 bins)	NFW preferred (ΔAIC > 160)	NFW adequate
ALFALFA $\times$ Yang BTFR	2,273 galaxies	Dense scatter larger at all W_{50}	Complex
Cluster-scale RAR	20 CLASH clusters	$a_0 \approx 17 g_\dagger$	Different regime
Kurtosis disambiguation	Tier K (250)	Spike driven by Korsaga B-band	Diagnostic
Kurtosis + MHONGOOSE	Tier K + 82 pts	κ_4 unchanged (20.7)	Robust to new data
TF z-evolution	Literature	$p = 0.045$ (systematics-dominated)	Inconclusive
TNG ξ mass-matched	131 SPARC–TNG pairs	Matched $C_{\rm TNG} = 0.13$, $p = 1.00$	Diagnostic
$M_{\rm dyn}$ triangulation	1,070 ALFALFA$\times$Yang + 57 SPARC	Estimator bias < 0.02 dex; TNG offset is inner concentration	Diagnostic

Table 6. Test battery scorecard.

Category	Count
BEC-consistent or supporting	10
Inconclusive or fragile	5
Failed / not replicated	1
Diagnostic (not directional)	4
Different regime (null expected)	3

NOTE—Supporting (10): environmental scatter, MC distance, non-parametric inversion, jackknife, split-half, structure-only null, extended RAR, Brouwer lensing, kurtosis disambiguation, kurtosis + MHONGOOSE. Inconclusive (5): environmental confound control, propensity matching, binning robustness (wide bins), ALFALFA BTFR, TF redshift evolution. Failed (1): PROBES replication. Diagnostic (4): kurtosis disambiguation, soliton-NFW composite, TNG ξ mass-matched, $M_{\rm dyn}$ triangulation. Different regime (3): cluster-scale RAR, cluster σ scaling, lensing profile shape.

7. SCORECARD AND INTERPRETATION

7.1. *Full Scorecard*

Table 6 presents an honest accounting of all 23 tests (16 core + 6 supporting).

7.2. *What Is Established*

Three results are robust:

1. The RAR is algebraically identical to the Bose-Einstein distribution (Equation 3). This is a mathematical fact.

2. RAR residual variance follows wave-interference scaling ($\sigma^2 \propto \bar{n}^2$), not Poisson ($\sigma^2 \propto \bar{n}$), with ΔAIC $= +10.5$.

3. The scatter inversion is anchored at $g_\dagger$ with sub-0.1 dex precision in the SPARC core sample, reproduced across 4 non-parametric methods, 25 binning configurations, 1,000 split-half trials (91.4% success), and 131 jackknife removals (std $= 0.005$ dex). It is absent in a structure-only semi-analytic null (offset by 1.7 dex).

7.3. *What Is Not Established*

1. **Quantum vs. classical.** The quantum prediction $\bar{n}(\bar{n}+1)$ is indistinguishable from the classical $\bar{n}^2$ (ΔAIC $= -0.79$). Approximately 4.75×10^6 data points are needed for a 3σ distinction.

2. ΛCDM **exclusion.** Our structure-only null fails to reproduce the features, but full-physics hydrodynamic simulations have not been tested with identical pipelines. We explicitly do not claim that ΛCDM is excluded.

3. **Kurtosis provenance.** The spike relies on Korsaga B-band mass models ($\kappa_4 = 23.95$); SPARC alone gives $\kappa_4 = 0.64$.

4. **Independent replication.** The PROBES catalog does not reproduce the scatter inversion. Whether this reflects data quality ($\sigma_{\rm PROBES} =$

0.32 vs. $\sigma_{\rm SPARC} = 0.17$) or the absence of a physical signal cannot be determined without homogeneous mass decomposition.

5. **Environmental scatter.** The null result is BEC-consistent but fragile: covariate imbalance (distance, inclination, gas fraction) between field and dense subsamples may mask a real difference.

7.4. *Three Possibilities*

The observations are consistent with three interpretations:

(a) **Coincidence**: the RAR's functional form happens to match the BE distribution, and the statistical features at $g_\dagger$ arise from the structure of SPARC galaxy mass profiles, disappearing in other catalogs (PROBES).

(b) ΛCDM **consequence**: baryonic feedback produces RAR statistics with BE-like structure, and $g_\dagger = cH_0/6$ arises from the cosmic baryon fraction.

(c) **Wave dark matter**: dark matter is a bosonic field whose collective dynamics produce BE occupation statistics, with $g_\dagger$ as a physical condensation threshold set by Λ, detectable only with high-quality mass decomposition.

Possibility (c) connects to the wave dark matter literature: Cheong et al. (2024) derive $\bar{n}(\bar{n}+1)$ fluctuations from the bosonic density matrix; Centers et al. (2021) confirm Rayleigh statistics in virialized FDM simulations; Khoury (2016) and Berezhiani & Khoury (2015) predict condensation in galaxy halos from superfluid dark matter. At cluster scales, Tian et al. (2020) find $a_0 = 1.73 \times 10^{-9}\,{\rm m\,s^{-2}} = 14.4 \times g_\dagger$, with a mass-dependent trend ($r = 0.58$, $p = 0.008$) consistent with a healing length $\xi \propto M^{-0.27}$.

No current test uniquely selects among these three possibilities.

8. PREDICTIONS AND FUTURE TESTS

1. **BIG-SPARC** (Haubner et al. 2024): $\sim$ 3,900 galaxies with SPARC-quality 3.6 μm mass decomposition. If the kurtosis spike persists at $g_\dagger$ with homogeneous photometry, the B-band artifact hypothesis is excluded. If the scatter inversion replicates, the PROBES failure is attributed to mass-model quality rather than absence of signal.

2. **Quantum correction**: Approximately 4.75×10^6 RAR data points needed for a 3σ distinction between $\bar{n}(\bar{n}+1)$ and $\bar{n}^2$.

3. **High-resolution lensing**: Euclid and HSC stacked profiles can resolve the 2–10 kpc solitonic core regime currently unresolved in KiDS (35 kpc inner radius).

4. **Hydrodynamic simulation comparison**: A preliminary healing-length comparison using 48,060 TNG100 halos (§6.7) shows that the apparent ξ-organizing signal vanishes under mass matching. The kurtosis spike and scatter inversion have not yet been tested in simulations with identical pipelines; this remains the most direct test of possibility (b).

5. **Stellar stream gaps**: Gaia DR3 gaps probe dark substructure fluctuations; wave dark matter predicts super-Poissonian spacing.

9. SUMMARY

The radial acceleration relation is algebraically identical to the Bose-Einstein occupation number, $g_{\rm DM}/g_{\rm bar} = 1/(e^{\sqrt{g_{\rm bar}/g_\dagger}} - 1)$. We have tested the statistical consequences of this mapping across six datasets spanning four quality tiers and three orders of magnitude in sample size.

Three fingerprints emerge in the SPARC core sample (131 galaxies): wave-interference variance scaling (ΔAIC $= +10.5$), a scatter inversion anchored at $g_\dagger$ to 0.05 dex precision, and a candidate kurtosis spike at $g_\dagger$ (Tier K only, B-band dependent). A battery of 23 tests yields 10 supporting, 5 inconclusive, 1 failure (PROBES), 4 diagnostic, and 3 in a different physical regime.

The PROBES replication failure is the strongest constraint: either the scatter inversion requires SPARC-quality mass decomposition to detect, or it is not a universal feature of galaxy rotation curves. BIG-SPARC ($\sim$ 3,900 galaxies, SPARC-quality photometry) is the decisive upcoming test.

DATA AND CODE AVAILABILITY

Analysis code, figure-generation scripts, all 23 summary JSON files, dataset metadata, and reproducibility documentation are available at the Open Science Framework: https://osf.io/jshr9/. The OSF project contains:

- `paper1.pdf` — compiled manuscript;
- `core/summary_*.json` — 16 core test result files;
- `supporting/summary_*.json` — 7 supporting test result files;
- `source/` — LaTeX source, bibliography, and `aastex631.cls`;

- `figures/` — publication figures (PDF);
- `MANIFEST.csv`, `CHECKSUMS.csv` — file inventory and SHA-256 checksums.

Minimum reproducer (after cloning the repository and acquiring SPARC data):

```
python3 analysis/tests/test_nonparametric_inversion.py
python3 analysis/tests/test_kurtosis_phase_transition.py
python3 analysis/tools/make_paper_assets.py
```

These three commands regenerate the scatter-inversion and kurtosis results (Figures 3–4) and all paper tables. The SPARC data are publicly available at http://astroweb.cwru.edu/SPARC/. The Korsaga et al. (2019) mass models are available via VizieR.

This research made use of the SPARC database (Lelli et al. 2016), the GHASP survey (Korsaga et al. 2019), the PROBES compilation (Stone et al. 2022), KiDS-1000 lensing data (Brouwer et al. 2021), and NASA/IPAC Extragalactic Database (NED).

Facilities: Spitzer, VLA, IRAM, Fabry-Perot

Software: NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), Matplotlib (Hunter 2007), statsmodels (Seabold & Perktold 2010)

APPENDIX

Table 7. Milgrom vs. standard RAR ΔAIC sensitivity.

Weighting scheme	ΔAIC
Uniform per-point	−1,222
Inverse-variance	≈ −400
Galaxy-averaged (1 per galaxy)	≈ −80

NOTE—All three schemes prefer the Milgrom form. The improvement is concentrated in the dark-matter-dominated regime ($\log g_{\rm bar} < -10$), where the two interpolation functions diverge.

A. AIC METHODOLOGY AND SENSITIVITY

Throughout this paper, we use the Akaike Information Criterion defined as

$$\mathrm{AIC} = \chi^2 + 2k, \tag{A1}$$

where k is the number of free parameters and

$$\chi^2 = \sum_{i=1}^{N} \left[\frac{\log g_{\mathrm{obs},i} - \log g_{\mathrm{model},i}}{\sigma_i} \right]^2 . \tag{A2}$$

For variance-scaling tests (§4.1), σ_i is the per-bin RMS scatter. For the Milgrom vs. standard RAR comparison (§3.3), σ_i is set uniformly to the global RMS under the default weighting.

Sensitivity of the Milgrom ΔAIC. The ΔAIC = −1,222 preference for the Milgrom de Sitter–Unruh form is obtained under uniform per-point weighting in $\log g_{\rm bar}$:

B. REPRODUCIBILITY

All results in this paper are derived from scripts in the project repository. To reproduce:

```
# Core fingerprints and robustness tests
python3 analysis/tests/test_*.py

# Milgrom vs standard RAR delta-AIC
# (Table 7 / Appendix A sensitivity)
python3 analysis/pipeline/\
  literature_crossref_tests.py

# Figures
python3 analysis/tools/make_paper_assets.py
```

Random seeds are fixed at 42 for all Monte Carlo, bootstrap, and permutation procedures. Summary JSON files for all 23 tests are archived at the OSF project (https://osf.io/jshr9/). Each JSON includes: test name, verdict, all numerical results, software versions, and the git SHA of the code that produced it.

REFERENCES

Berezhiani, L., & Khoury, J. 2015, Physical Review D, 92, 103510, doi: 10.1103/PhysRevD.92.103510

Brouwer, M. M., et al. 2021, Astronomy & Astrophysics, 650, A113, doi: 10.1051/0004-6361/202040108

Centers, G. P., et al. 2021, Nature Communications, 12, 7321, doi: 10.1038/s41467-021-27632-7

Cheong, D. Y., Rodd, N. L., & Wang, L.-T. 2024, arXiv e-prints. https://arxiv.org/abs/2408.04696

de Blok, W. J. G., & Bosma, A. 2002, Astronomy & Astrophysics, 385, 816, doi: 10.1051/0004-6361:20020080

Ferreira, E. G. M. 2021, The Astronomy and Astrophysics Review, 29, 7, doi: 10.1007/s00159-021-00135-6

Harris, C. R., et al. 2020, Nature, 585, 357, doi: 10.1038/s41586-020-2649-2

Haubner, K., Lelli, F., Di Teodoro, E., et al. 2024, arXiv e-prints. https://arxiv.org/abs/2411.13329

Hunter, J. D. 2007, Computing in Science & Engineering, 9, 90, doi: 10.1109/MCSE.2007.55

Keller, B. W., & Wadsley, J. W. 2017, The Astrophysical Journal Letters, 835, L17, doi: 10.3847/2041-8213/835/1/L17

Khoury, J. 2016, Physical Review D, 93, 103533, doi: 10.1103/PhysRevD.93.103533

Korsaga, M., Carignan, C., Amram, P., Epinat, B., & Jarrett, T. H. 2019, Monthly Notices of the Royal Astronomical Society, 490, 2977, doi: 10.1093/mnras/stz2678

Lelli, F., McGaugh, S. S., & Schombert, J. M. 2016, The Astronomical Journal, 152, 157, doi: 10.3847/0004-6256/152/6/157

Lelli, F., McGaugh, S. S., Schombert, J. M., & Pawlowski, M. S. 2017, The Astrophysical Journal, 836, 152, doi: 10.3847/1538-4357/836/2/152

Li, P., Lelli, F., McGaugh, S., & Schombert, J. 2018, Astronomy & Astrophysics, 615, A3, doi: 10.1051/0004-6361/201732547

Ludlow, A. D., et al. 2017, Physical Review Letters, 118, 161103, doi: 10.1103/PhysRevLett.118.161103

McGaugh, S. S., Lelli, F., & Schombert, J. M. 2016, Physical Review Letters, 117, 201101, doi: 10.1103/PhysRevLett.117.201101

Milgrom, M. 1983, The Astrophysical Journal, 270, 365, doi: 10.1086/161130

—. 1999, Physics Letters A, 253, 273, doi: 10.1016/S0375-9601(99)00077-8

Riess, A. G., et al. 2022, The Astrophysical Journal Letters, 934, L7, doi: 10.3847/2041-8213/ac5c5b

Sanders, R. H. 2008, Monthly Notices of the Royal Astronomical Society, 386, 1588, doi: 10.1111/j.1365-2966.2008.13140.x

Schive, H.-Y., Chiueh, T., & Broadhurst, T. 2014, Nature Physics, 10, 496, doi: 10.1038/nphys2996

Seabold, S., & Perktold, J. 2010, in 9th Python in Science Conference, 92–96, doi: 10.25080/Majora-92bf1922-011

Stone, C. J., Courteau, S., Cuillandre, J.-C., et al. 2022, The Astrophysical Journal Supplement Series, 259, 44, doi: 10.3847/1538-4365/ac83ad

Tian, Y., Umetsu, K., Ko, C.-M., Donahue, M., & Morandi, A. 2020, The Astrophysical Journal, 896, 70, doi: 10.3847/1538-4357/ab8e3d

Verlinde, E. P. 2017, SciPost Physics, 2, 016, doi: 10.21468/SciPostPhys.2.3.016

Virtanen, P., et al. 2020, Nature Methods, 17, 261, doi: 10.1038/s41592-019-0686-2

Yang, X., Mo, H. J., van den Bosch, F. C., et al. 2007, The Astrophysical Journal, 671, 153, doi: 10.1086/522027